BLUESTOCK FINTECH

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Intern Capstone Project | Bluestock Fintech Pvt. Ltd.

June 2026 | 7 Working Days | ~50 Hours

1. Executive Summary

This capstone project delivers a full-stack Mutual Fund Analytics Platform for Bluestock Fintech Pvt. Ltd., built over 7 working days. The platform ingests publicly available AMFI India mutual fund data, transforms it through a robust ETL pipeline, stores it in a relational SQLite database with a 7-table star schema, performs comprehensive exploratory data analysis and performance analytics, and presents insights via an interactive 5-page Power BI dashboard.

The project analyzed 40 mutual fund schemes across 10 fund houses, covering ~46,000 daily NAV records and ~32,778 investor transactions spanning January 2022 to May 2026. Key deliverables include automated Python scripts, a verified SQL database, 14 EDA charts, computed risk metrics (VaR, Sharpe, Drawdown), a fund scorecard, and an interactive BI dashboard.

Three significant dataset integrity findings emerged from cross-validation work: (1) Sharpe/Sortino ratios are mathematically unstable for Liquid funds near the risk-free rate; (2) benchmark indices were generated independently of NAV data, making Alpha/Beta calculations non-meaningful; and (3) provided Max Drawdown figures were inconsistent with actual NAV series. All findings are documented transparently with evidence and corrective decisions taken.

2. Data Sources & Datasets

2.1 Dataset Overview

All datasets are derived from publicly available AMFI India, NSE, BSE, and mfapi.in data sources. No proprietary or confidential data was used.

Dataset	Rows	Description
01_fund_master.csv	40	40 real mutual fund schemes (AMFI codes, fund house, category)
02_nav_history.csv	46,000	Daily NAV for all 40 schemes, Jan 2022 – May 2026
03_aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses, 2022–2025
04_monthly_sip_inflows.csv	48	Monthly SIP inflow, accounts, AUM (real AMFI data)
05_category_inflows.csv	144	Net inflows by fund category, FY 2024–25
06_industry_folio_count.csv	21	Total MF folios growth, Jan 2022 – Dec 2025
07_scheme_performance.csv	40	Risk/return metrics per scheme
08_investor_transactions.csv	32,778	SIP/Lumpsum/Redemption transactions, 5,000 investors

09_portfolio_holdings.csv	322	Equity holdings, sector weights, Dec 2025
10_benchmark_indices.csv	8,050	Daily NIFTY50, NIFTY100, and other indices

2.2 Live Data

Live NAV data was additionally fetched from the mfapi.in REST API for 6 key schemes (HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, Kotak Bluechip) using a retry/backoff-enabled Python script. This provided historical data going back to 2012–2013 for some schemes.

3. System Architecture & ETL Pipeline

3.1 Pipeline Overview

The project follows a classic 5-layer data engineering architecture:

- Layer 1 — Extract: Raw CSVs from AMFI data + live API calls to mfapi.in
- Layer 2 — Transform: Python (Pandas) cleaning, validation, date parsing, derived fields
- Layer 3 — Load: SQLite database with 7-table star schema via SQLAlchemy
- Layer 4 — Analyse: Jupyter notebooks for EDA and performance metrics
- Layer 5 — Visualise: Power BI Desktop 5-page interactive dashboard

3.2 Database Schema

A 7-table star schema was designed in SQLite. dim_fund (40 rows) serves as the central dimension table, connected via amfi_code foreign keys to 5 fact tables: fact_nav (46,000 rows), fact_transactions (32,778 rows), fact_performance (40 rows), fact_aum (90 rows), and fact_sip_industry (48 rows). A dim_date calendar table (5,479 rows covering 2012–2026) supports date-based aggregations.

All row counts were verified post-load via SELECT COUNT(*) queries, confirming a 100% match between source CSVs and loaded database tables across all 7 tables.

3.3 Data Quality Findings

Key findings from data validation:

- 12 null values in yoy_growth_pct (monthly_sip_inflows) — expected, as the first 12 months have no prior-year baseline for YoY computation
- All date columns loaded as string dtype by default — parsed to datetime during Day 2 cleaning
- 0 duplicate rows across all 10 datasets
- 40/40 AMFI codes validated between fund_master and nav_history (100% match)
- scheme_performance.csv contains Equity and Debt categories only — no Hybrid funds present, despite the project specification mentioning a third category
- risk_category has 5 distinct values (Low, Moderate, Moderately High, High, Very High) vs the 4 listed in the appendix

4. Exploratory Data Analysis — Key Findings

14 charts were produced across 9 analysis areas. The 10 most significant findings are:

- 2023 was a clear bull run and 2024 showed a visible correction across nearly all 40 schemes' NAV trends
- SBI Mutual Fund dominates industry AUM at Rs.12.50 lakh crore (Dec 2025), confirmed against real AMFI data
- Monthly SIP inflows hit Rs.31,002 crore (Dec 2025), the peak of a consistent upward trend since Jan 2022
- Every fund category recorded positive net inflows every month in FY 2024–25 — no category saw net outflows
- Transaction frequency is broadly consistent across all age groups — no demographic transacts disproportionately more or less
- SIP amounts show a bimodal distribution (gap between Rs.30,000–40,000) not explained by income or city tier
- Madhya Pradesh leads in total SIP amount while Punjab leads in transaction count
- Total MF folios grew from 13.26 crore (Jan 2022) to 26.12 crore (Dec 2025), crossing 20 crore in Oct 2024
- Debt funds show near-zero correlation (-0.07 to +0.05) with Equity funds, confirming genuine diversification value
- Banking (19.2%) and IT are the two largest sectors across all equity fund holdings

5. Performance Analytics

5.1 Metrics Computed

The following metrics were computed from raw NAV history for all 40 funds:

Metric	Method	Source Used
CAGR (1yr, 3yr, 5yr)	(NAV_end/NAV_start)^(1/actual_years) - 1	Calculated from nav_history
Sharpe Ratio	(Rp - Rf) / Std(Rp) × √252, Rf=6.5%	Provided (scheme_performance.csv)
Sortino Ratio	Sharpe variant using downside std only	Provided (scheme_performance.csv)
Alpha / Beta	OLS regression vs NIFTY100	Provided (see note below)
Max Drawdown	min(NAV / running_max - 1)	Calculated from nav_history
VaR (95%)	5th percentile of daily return distribution	Calculated from nav_history
CVaR (95%)	Mean of returns below VaR threshold	Calculated from nav_history

5.2 Dataset Integrity Findings

Three significant discrepancies were discovered through systematic cross-validation of computed metrics against provided values:

Finding 1 — Sharpe/Sortino instability for Liquid funds: ICICI Pru Liquid Fund shows annualised return of 6.74% against a 6.5% risk-free rate, producing a near-zero excess return. Since the Sharpe denominator (volatility) is also extremely small for Liquid funds (~0.49% annualised), the ratio becomes mathematically unstable. The provided Sharpe of 7.68 and our calculated value near zero both reflect this instability — we used the provided values for all downstream calculations.

Finding 2 — Benchmark independence: OLS regression of fund returns against NIFTY100 produced near-zero Beta values (-0.07 to +0.10) and average |r| = 0.02 across all 40 funds, including index funds. Raw NAV and benchmark close values were visually confirmed to move in opposite directions on the same trading days — confirming nav_history.csv and benchmark_indices.csv were generated as statistically independent series. The provided Alpha/Beta values were used instead.

Finding 3 — Max Drawdown discrepancies: Calculated values showed 15–37 percentage point gaps vs provided values, with no consistent direction. For Axis Small Cap Fund, calculated drawdown of -51.68% was directly verified against the NAV chart (visually confirmed 50%+ peak-to-trough decline). Calculated values were used as they reflect the actual NAV series.

5.3 Fund Scorecard

A composite 0–100 scorecard was built using percentile ranks weighted per project specifications: 30% × 3yr return + 25% × Sharpe + 20% × Alpha + 15% × expense ratio (inverse) + 10% × max drawdown (inverse).

Top 3 performing funds by composite score: (1) Mirae Asset Large Cap Fund — Regular — Growth (73.4), (2) Kotak Flexicap Fund — Regular — Growth (72.8), (3) ICICI Pru Bluechip Fund — Direct — Growth (67.4).

Note: Two Debt/Liquid funds appear in the top 5, driven by inflated Sharpe ratios from the instability documented above. The scorecard should be interpreted with category context rather than as a universal ranking.

6. Interactive Dashboard — Power BI

A 5-page interactive Power BI dashboard was built connecting to all cleaned datasets. All pages include at least 2 interactive slicers per project requirements.

Page 1 — Industry Overview

KPI cards: Total Industry AUM (Rs.81L Cr), Peak SIP Inflow (Rs.31,002 Cr), Latest Folios (26.12 Cr), Schemes in dataset (40). Line chart: industry AUM trend 2022–2025. Bar chart: AUM by fund house, filtered to latest available period.

Page 2 — Fund Performance

Scatter plot: 3yr return vs annualised std dev, bubble size = AUM, colored by category. Sortable fund scorecard table. Dual-axis NAV vs benchmark chart (illustrative only — see dataset integrity notes). Slicers: fund house, category, plan.

Page 3 — Investor Analytics

Bar chart: total transaction amount by state. Donut: SIP/Lumpsum/Redemption split. Bar: average SIP amount by age group (DAX measure, SIP transactions only). Monthly transaction volume line chart. Slicers: state, age group, city tier.

Page 4 — SIP & Market Trends

Dual-axis line and column chart: SIP inflow (bars) + NIFTY50 (line) 2022–2025. Matrix heatmap: category net inflows by month with conditional background formatting. Bar chart: top 5 categories by net inflow.

Page 5 — NAV Detail (Drill-through)

Drill-through page accessible from the Fund Performance scorecard table. Shows fund-specific NAV history line chart and independent benchmark index chart for reference.

7. Advanced Analytics

7.1 Value at Risk (VaR) and CVaR

Historical VaR at 95% confidence was computed for all 40 funds. Small Cap funds dominate the worst-VaR list: SBI Small Cap Direct shows the worst daily tail loss at -2.69% VaR / -3.24% CVaR. CVaR was consistently more negative than VaR for every fund, confirming formula correctness.

7.2 Rolling 90-Day Sharpe Ratio

No fund maintained a consistently positive 90-day rolling Sharpe throughout 2022–2026. All 5 selected funds oscillated between positive and negative values, with SBI Magnum Gilt Fund showing the most extreme swings (below -4), consistent with Liquid/Debt fund Sharpe instability.

7.3 Investor Cohort Analysis

Two cohorts identified: 2024 (4,803 investors, avg SIP Rs.10,997) and 2025 (197 investors, avg SIP Rs.13,505). The 2025 cohort invests 23% more per SIP, though sample size (197) warrants cautious interpretation.

7.4 SIP Continuity

97.8% of investors with 6+ SIP transactions show average gaps exceeding 35 days (mean: 64.89 days, roughly bi-monthly). This irregularity is demographically uniform — no single age group shows disproportionate at-risk rates.

7.5 Fund Recommender

A rule-based recommender (scripts/recommender.py) returns top 3 funds by Sharpe ratio within risk appetite mapping: Low → Liquid funds (Sharpe 5–8, inflated), Moderate → Large Cap equity (Sharpe ~1.06, Rs.14–15% 3yr return), High → Mid/Small Cap equity (Sharpe ~0.94–0.96, Rs.18–23% 3yr return).

7.6 Sector HHI Concentration

HHI ranged from 1,240 (UTI Mid Cap — most diversified) to 2,968 (Axis Bluechip — most concentrated). Counterintuitively, Small Cap funds scored lower HHI (more diversified) than several Large Cap and Mid Cap funds, likely due to broader mandatory stock selection across smaller companies.

8. Recommendations

• For moderate-risk investors seeking growth: Mirae Asset Large Cap Fund (composite score 73.4) and Kotak Flexicap Fund (72.8) offer strong risk-adjusted returns with reasonable expense ratios

• For high-risk investors: SBI Small Cap Regular Plan offers the highest 3yr CAGR (23.39%) but carries the worst max drawdown (-28.7%) and highest daily VaR (-2.45%) — suitable only for long-horizon investors

• Portfolio diversification: combining any Debt fund with Equity funds provides genuine diversification (correlation -0.07 to +0.05), not merely a label difference

• SIP discipline: the 97.8% at-risk rate suggests a platform intervention (reminders, automated top-up) could significantly improve SIP continuity and investor outcomes

• Dashboard refresh: the ETL pipeline can be re-run weekly to refresh Power BI with latest NAV data from mfapi.in, requiring only python scripts/etl_pipeline.py to execute

9. Limitations & Future Work

• Benchmark independence: Alpha/Beta metrics are illustrative only due to statistically independent NAV and benchmark generation in this dataset

• CAGR '5yr' figure uses ~4.4 years of actual data — true 5-year CAGR requires data going back to 2021

• Investor transaction data is synthetically generated — demographic and geographic patterns may not precisely reflect real AMFI investor distributions

• Sharpe ratio instability for Liquid/Debt funds is an inherent formula limitation when fund return approaches the risk-free rate, not a project-specific bug

• Future work: Monte Carlo NAV projection, Markowitz Efficient Frontier optimisation, automated weekly email reporting, Streamlit web dashboard

10. Technical Stack

Category	Tool / Library	Purpose
Language	Python 3.13	All data processing, ETL, analytics
Data Manipulation	Pandas 2.x	DataFrames, cleaning, aggregation
Numerical	NumPy 1.x	Risk metrics, statistical functions
Statistics	SciPy	OLS regression for Beta/Alpha
Visualization	Matplotlib, Seaborn, Plotly	Static and interactive charts
Database	SQLite3 + SQLAlchemy 2.0	Relational database + Python ORM
Notebooks	Jupyter Lab	EDA and analytics
Dashboard	Power BI Desktop	Interactive BI dashboard
API	mfapi.in	Live NAV data (no auth required)
Version Control	Git + GitHub	Code versioning, submission